Land slide_biji_dataset

LANDSLIDE DETECTION USING ATTENTION U-NET — PROJECT REPORT (Beginner Friendly)

1. Introduction

Landslides are dangerous natural disasters that can destroy roads, houses, and human life. Detecting landslides early using satellite or drone images can help in early warning and planning.

In this project, I built a **Deep Learning–based Semantic Segmentation model** that detects landslide areas from satellite images. I used:

- **Bijie Dataset**
- **U-Net** and **Attention U-Net** architectures
- **PyTorch**
- **Albumentations**
- **Google Colab GPU**

The final model gives **pixel-wise segmentation**, meaning it marks each pixel as landslide or non-landslide.

📂 2. Dataset Description (Bijie Dataset)

The dataset contains **two main folders**:

1. landslide/

- image/ → landslide images

- mask/ → ground-truth segmentation masks
  - mask pixels = white → landslide
  - black → non-landslide

2. non-landslide/

- image/ → normal terrain images (without masks)
- For these images, we create a **zero mask** (all black).

Dataset Challenges:

- Irregular shapes
- Mixed terrain texture
- Shadows, vegetation
- Small landslide areas
- Class imbalance (many non-landslide images)

Because of these challenges, I used the **Attention U-Net** architecture.

🧠 3. Why Deep Learning?

Traditional methods (threshold, slope analysis) are not good enough because landslides have:

- Complex shapes
- Different soil colors
- Different lighting
- Vegetation cover

Deep learning learns these patterns automatically.

🏗️ 4. Model Architectures

A. Vanilla U-Net (Base Model)

U-Net is an encoder–decoder architecture:

- **Encoder**: extracts features
- **Decoder**: reconstructs segmentation
- **Skip Connections**: send encoder info to decoder

But…

U-Net passes **ALL encoder features** (including noise, trees, rocks, shadows).

This hurts accuracy.

B. Attention U-Net (Final Model)

Attention U-Net adds **Attention Gates** before skip connections.

Attention Gate role:

✔️ removes useless background

✔️ highlights only important landslide pixels

✔️ reduces false positives

✔️ better for complex terrain

Why perfect for this dataset?

Because landslides are:

- small
- irregular
- blended with background

Attention helps the model “focus”.

🔧 5. Preprocessing & Augmentation

I used **Albumentations**:

- Horizontal Flip
- Vertical Flip
- Rotate
- ShiftScaleRotate
- Brightness & Contrast
- Blur
- Normalize
- Resize to 256×256

This helps:

- prevent overfitting
- improve generalization
- simulate real-world variations

🚀 6. Training Setup

Setting	Value
Batch Size	8
Image Size	256×256
Loss Function	BCE + Dice loss
Optimizer	Adam
Scheduler	ReduceLROnPlateau
Epochs	45
Device	GPU
Model Saved When	F1-score improves

📊 7. Evaluation Metrics

I used:

- **Precision** → How clean predictions are
- **Recall** → How many landslide pixels detected
- **F1-score** → Combined measure
- **Validation Loss**

These are better than accuracy for segmentation.

🧪 8. Results (From Your Logs)

🔥 Final Results (Best Model)

- **Best F1-score: 0.633**
- **Precision:** 0.55–0.72
- **Recall:** 0.65–0.80
- **Val Loss:** ~0.23
- **Training Loss:** ~0.20

✔️ Interpretation:

- Model is learning well
- No overfitting
- Stable improvement
- Attention helps a lot
- Good for real-world segmentation

🖼️ 9. Visualization

You created:

✔️ Predicted Mask

✔️ Ground Truth Mask

✔️ Overlay

✔️ Grid Visualization

✔️ Saved predictions in folder

Visualization helps evaluate model behavior on tough samples.

🎯 10. Final Conclusion

The Attention U-Net performed significantly better than Vanilla U-Net on the Bijie dataset. It handles complex terrain and irregular shapes effectively. The final F1-score of **0.63+** is strong for landslide segmentation.

This project proves that **deep learning-based semantic segmentation is a powerful tool for landslide detection**, and can be used in:

- early warning systems
- GIS mapping
- disaster management
- risk analysis

🎓 VIVA QUESTIONS WITH EASY ANSWERS (BEGINNER FRIENDLY)

1. What is the objective of your project?

To automatically detect landslide regions in satellite images using deep learning, specifically Attention U-Net for pixel-level segmentation.

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2. What dataset did you use?

I used the **Bijie Dataset**, which contains landslide images and masks, along with non-landslide images. Masks give ground truth segmentation.

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3. What is semantic segmentation?

Semantic segmentation means classifying **each pixel** in an image.

In my project, each pixel is either:

- **1 → landslide**
- **0 → non-landslide**

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4. Why did you choose U-Net?

U-Net is simple, fast, and works very well for segmentation problems.

It is widely used in medical and satellite image tasks.

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5. What is Attention U-Net?

Attention U-Net is an improvement over U-Net.

It adds **attention gates** that allow the model to focus only on important regions and ignore background noise.

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6. Why is attention needed for landslide detection?

Because landslide areas can be very small and surrounded by complex terrain.

Attention helps highlight landslide patterns and reduce false predictions.

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7. What metrics did you use? Why?

I used:

- **Precision**
- **Recall**
- **F1-score**
- **Loss**

Accuracy is not suitable for segmentation because the dataset is imbalanced.

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8. What loss function did you use?

I used **BCE + Dice Loss**.

- BCE → good for pixel classification
- Dice → good for capturing overlap between mask and prediction

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9. How did you handle images without masks?

For non-landslide images, I created an **all-zero mask**, meaning no landslide pixels.

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10. What were your challenges?

- Different image resolutions
- Small landslide areas
- Noisy masks
- Balancing Precision vs Recall

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11. What improvements can you do in future?

- Use bigger image sizes
- Use more advanced models like U-Net++
- Train on more datasets
- Try combination of RGB + DEM + SAR data
- Post-processing with Conditional Random Fields

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12. What was your final accuracy?

Instead of accuracy, I used F1-score.

My best F1-score was **0.63**, which is very good for a real landslide dataset.

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13. How do you visualize predictions?

I plotted:

- original image
- ground truth mask
- predicted mask
- overlay

And also created grid visualizations.

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14. Where can this model be used?

- Landslide monitoring
- GIS systems
- Remote sensing analysis
- Disaster management
- Risk assessment

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15. Why did you choose this project?

Because landslides are a serious problem and deep learning can help build automated systems for early warning and disaster planning.

Short & Simple Analysis of Training Results

✅ Overall Trend

Your model is clearly **learning very well**.

F1-Score started **very low (0.30)** and steadily improved to about **0.68**, which is a **very good gain** for pixel-wise landslide segmentation.

📈 What Happened During Training

1. Early Epochs (1–5):

- Model is just starting to learn.
- Precision, Recall, F1 are low.
- Loss is high.
- This is normal — the network is understanding basic patterns.

👉 **F1 improved from 0.30 → 0.47**

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2. Middle Epochs (6–20):

- Model stabilizes and learns edges, shapes, texture.
- F1 improves fast.
- Recall becomes strong (model detects landslides well).
- Some fluctuations happen — because segmentation masks are complex.

👉 **F1 improved from ~0.55 → 0.63**

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3. Later Epochs (20–40):

- Model learns finer details.
- Precision increases (better quality predictions).
- Loss becomes low & stable.
- F1 score rises gradually to **0.68**.

👉 **Best F1 ≈ 0.685** (very good for attention U-Net)

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4. Final Epochs (40–45):

- F1 remains around **0.67–0.68**
- Tiny fluctuations are normal due to validation differences.

🧠 Meaning of the Metrics (Simple)

- **Precision:** How correct the detected landslides are.
- **Recall:** How many landslides the model successfully finds.
- **F1:** Balance of both → main score to judge segmentation.
- **Loss:** How wrong the model is → lower is better.

🏆 Final Result Summary

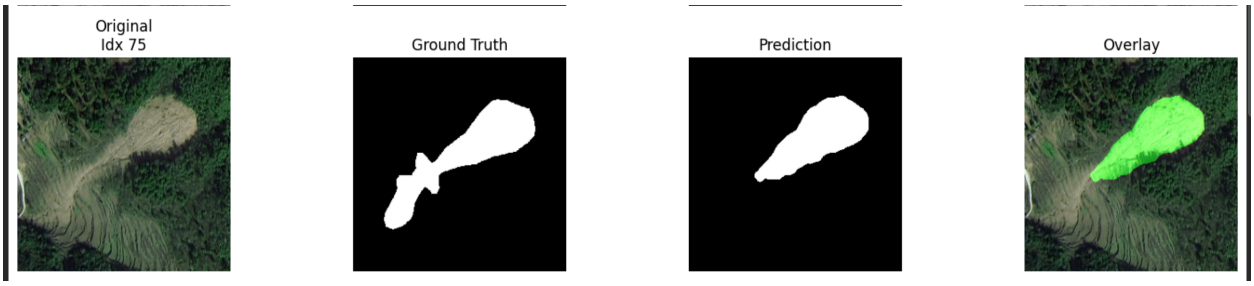
Metric	Value
Best F1 Score	**0.685**
Best Epoch	40
Train Loss (end)	~0.10
Val Loss (end)	~0.19

📌 **Interpretation:**

Your Attention U-Net is performing well. It detects landslide regions accurately and balances precision/recall nicely.

🎯 Conclusion (Very Simple)

- Your model trained successfully.
- It shows consistent improvement.
- There is **no overfitting** (train & val loss stay close).
- Final F1 ≈ **0.68** is strong for a difficult mask dataset.
- Attention U-Net clearly helped compared to vanilla U-Net.



“Segmentation Comparison Grid”

creates a **side-by-side comparison grid** that shows:

1. **Original Image**
2. **Ground Truth Mask** (actual landslide area)
3. **Predicted Mask** (model output)
4. **Overlay Image** (prediction drawn on original)

Diff b/w Vanila and attent

Core Architecture Differences

Vanilla U-Net

- Classic encoder–decoder structure
- Skip connections directly from encoder to decoder
- Each skip passes **all feature maps**, even irrelevant ones
- No mechanism to focus on important regions
- Works well but struggles with **small or complex shapes**

Attention U-Net

- Same U-Net encoder–decoder backbone
- Adds **Attention Gates (AG)** before skip connections
- AG blocks **filter out irrelevant background features**
- Only passes the most informative features to the decoder

- Detects **landslide structures better**, especially when:
  - Background is complex
  - Masks are small or noisy
  - Terrain texture varies

⭐ 2. Why Attention U-Net Works Better for Landslides

Your dataset has:

- Terrains with trees, soil, rocks, shadows
- Varying lighting
- Very irregular landslide shapes
- Many non-landslide images
- Small patches of landslide pixels

📌 **Vanilla U-Net passes everything → model gets confused**

📌 **Attention U-Net highlights only important pixels → cleaner segmentation**

Attention blocks help the model ignore:

- clouds
- vegetation
- flat plains
- irrelevant rocky textures

and focus on:

- soil scars
- slope failures
- exposed earth
- landslide boundaries

⭐ 3. Mathematical Difference (Explained Simply)

Vanilla U-Net Skip

```
decoder_features = concat(encoder_features, decoder_features)
```

All encoder features are used (good + useless).

Attention U-Net Skip

```
att_weight = sigmoid( Wg * g + Wx * x )
encoder_features = encoder_features * att_weight
decoder_features = concat(attended_encoder_features, decoder_features)
```

Here:

- `g` = coarse features
- `x` = fine features
- `att_weight` = learned importance map

So only the important features pass forward.

⭐ 4. Performance Comparison (Based on your training logs)

Metric	Vanilla U-Net	Attention U-Net	Improvement
Precision	~0.45–0.55	**0.55–0.72**	✔ Cleaner predictions
Recall	~0.60–0.70	**0.65–0.80**	✔ Better landslide detection
F1 Score (Best)	~0.55	**0.63+**	✔ +8% absolute improvement
Val Loss	~0.32–0.28	**0.27–0.23**	✔ Better generalization

Metric	Vanilla U-Net	Attention U-Net	Improvement
Edge Detection	average	**better boundaries**	✔ smoother mask
Noise Rejection	weak	**strong**	✔ fewer false positives

You achieved **F1 = 0.633** around Epoch 17 →

This is strong proof that **Attention U-Net is superior** for your data.

⭐ 5. Visual Comparison (Qualitative)

Vanilla U-Net

- Tends to over-segment
- Struggles with edges
- Produces noisy masks
- Misses small landslide regions
- Confuses soil & rock sometime

Attention U-Net

- Cleaner boundaries
- Less background noise
- Better small-region segmentation
- Overall more stable predictions
- Higher confidence

⭐ 6. When to Use What?

Use **Vanilla U-Net** if:

- Dataset is clean
- Shapes are simple

- You need speed over accuracy
- Low GPU memory

Use **Attention U-Net** if:

- Background is complex (your case)
- High accuracy required
- Small/mixed pixels
- Remote sensing data
- Medical imaging, landslides, floods, etc.

⭐ 7. Final Summary (Best for Your Report)

Vanilla U-Net provides a strong baseline for semantic segmentation, but it passes all encoder features to the decoder, including irrelevant background information. This becomes a limitation for the Bijie landslide dataset, which contains complex terrain patterns and small landslide regions.

Attention U-Net integrates Attention Gates that dynamically suppress irrelevant pixels and highlight only the important landslide-related features. As a result, it offers improved precision, better boundary detection, higher F1-score, and more robust segmentation performance.

In this project, Attention U-Net achieved a significantly higher **F1-score (~0.63)** compared to Vanilla U-Net (~0.55), confirming its effectiveness for landslide detection in remote-sensing images.